Polar Notation

Complex numbers can also be expressed in *polar notation*, besides the *rectangular notation* just described. For example, Fig. 30-2 shows three complex numbers in polar form, the same ones previously presented in Fig. 30-1. The **magnitude** is the length of the vector starting at the origin and ending at the complex point, while the **phase angle** is measured between this vector and the positive x-axis. Complex numbers can be converted between rectangular and polar notation by the following equations (paying attention to the polar notation *nuisances* discussed in Chapter 8):

EQUATION 30-8
Rectangular-to-polar conversion. The complex variable, A, can be changed from rectangular form: *Re* A & *Im* A, to polar form: *M* & θ.

$$M = \sqrt{(Re\ \mathrm{A})^2 + (Im\ \mathrm{A})^2}$$

$$\theta = \arctan\left[\frac{Im\ \mathrm{A}}{Re\ \mathrm{A}}\right]$$

EQUATION 30-9
Polar-to-rectangular conversion. This is changing the complex number from *M* & θ to *Re* A & *Im* A.

$$Re\ \mathrm{A} = M\cos(\theta)$$

$$Im\ \mathrm{A} = M\sin(\theta)$$

This brings up a giant leap in the mathematics. (Yes, this means you should pay extra attention). A complex number written in rectangular notation

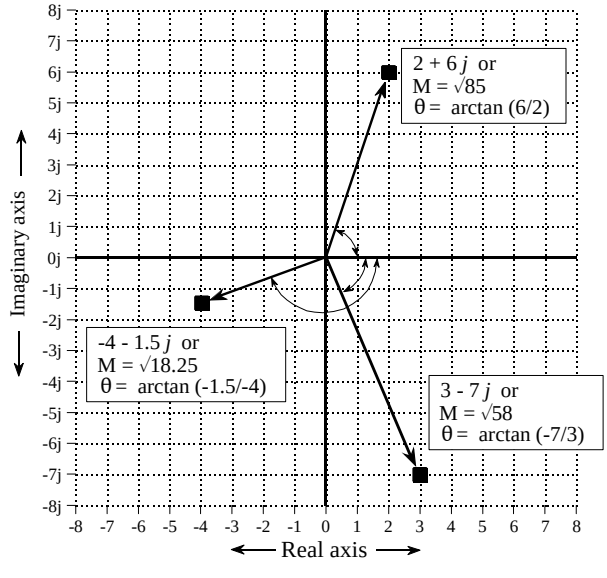

FIGURE 30-2
Complex numbers in polar form. Three example points in the complex plane are shown in polar coordinates. Figure 30-1 shows these same points in rectangular form.